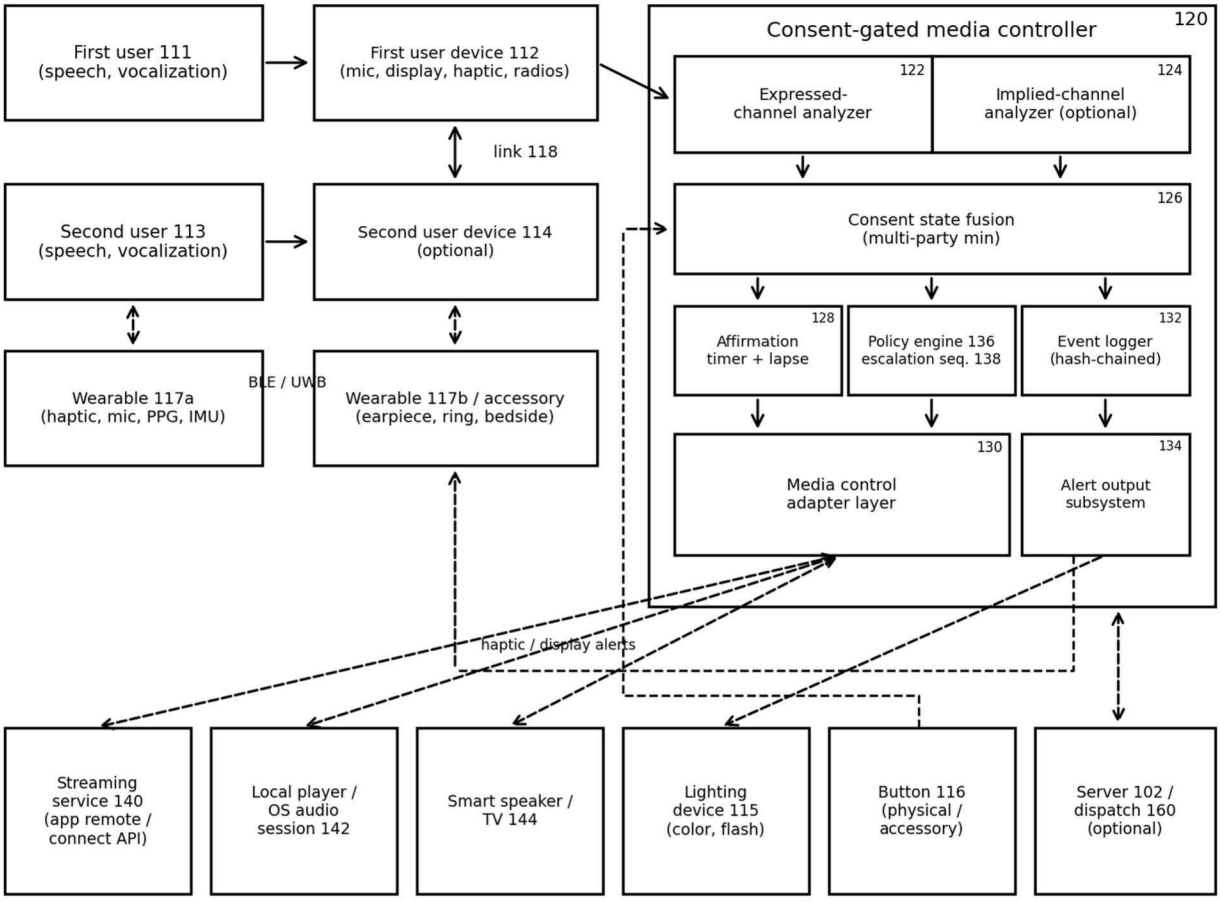
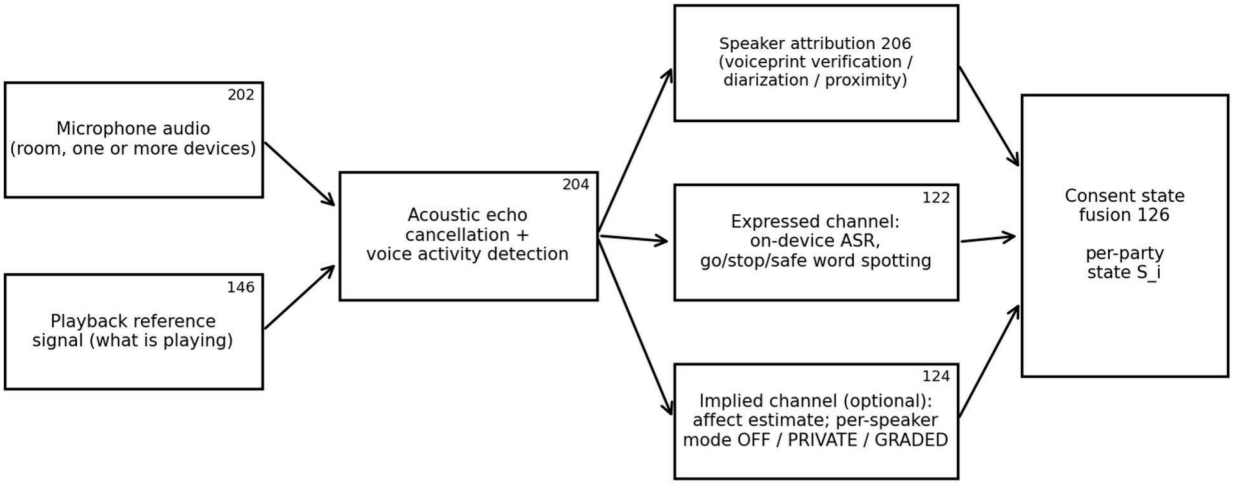
FIG. 1



146: playback reference (track, position, volume) fed back to 122/124 for echo cancellation

FIG. 2



210

Fusion rules

R1 STOP word or safe word from any party -> S = WITHDRAWN immediately (overrides everything)

R2 GO word at its enrolled tier -> S = AFFIRMED or ENTHUSIASTIC; refresh timer if tier meets policy

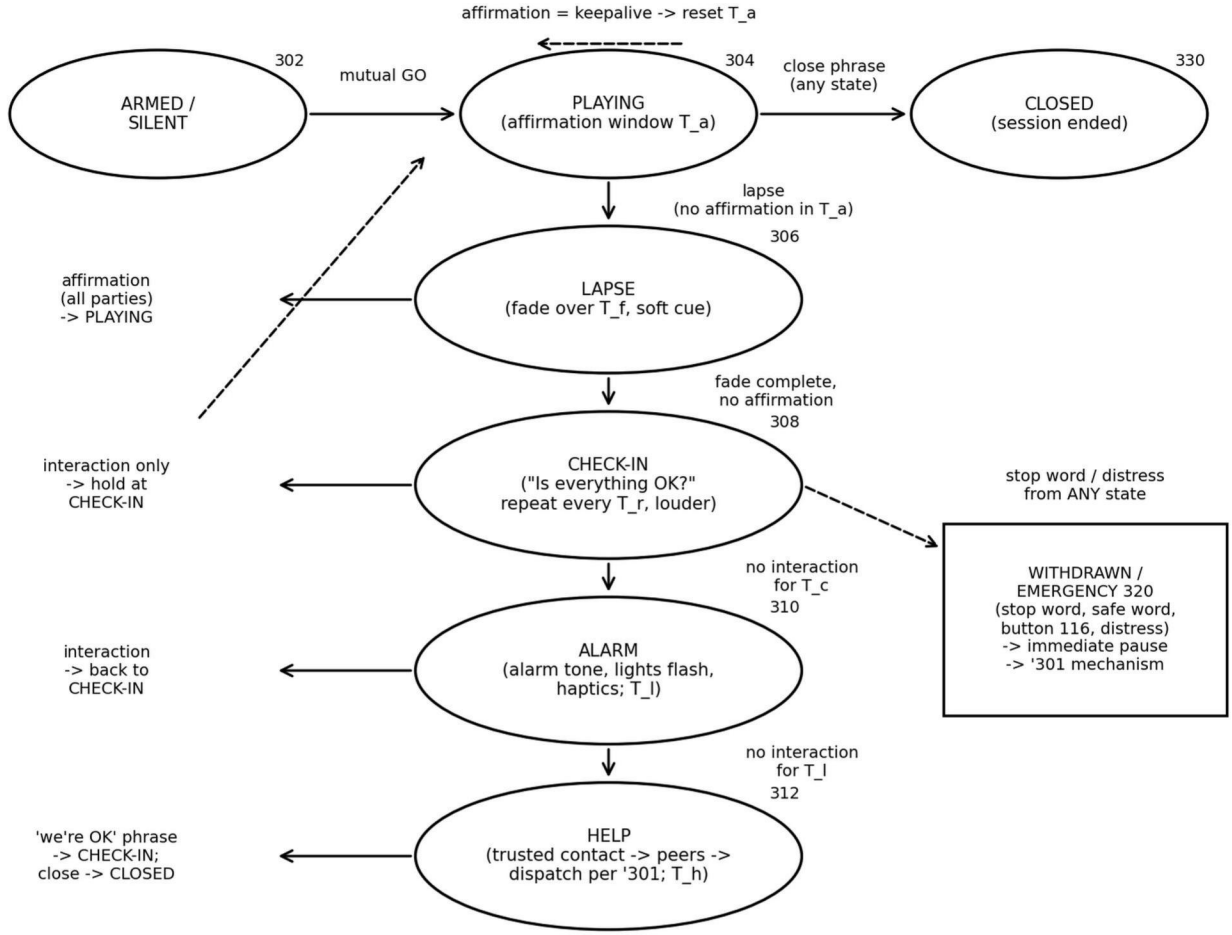
R3 per speaker's tone mode: OFF = skip; PRIVATE = own distress -> private prompt; GRADED = discordant GO -> AMBIGUOUS, no refresh

R4 Positive affect alone (no GO word) -> refresh only if 'implied refresh' enabled and within limit N

R5 Distress signal (vocal, biometric, button 116) -> S = WITHDRAWN; escalate per '301 emergency mechanism

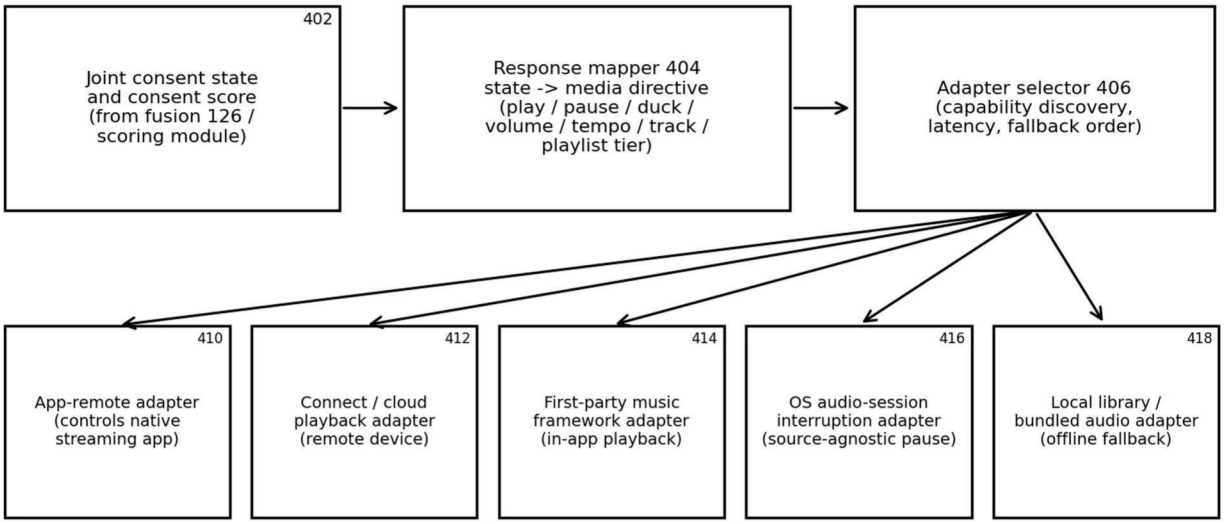
R6 Multi-party: S_joint = min(S_1, S_2, ... S_n) (lowest state governs)

FIG. 3



Interaction (speech, motion, touch, button) halts escalation but does not resume music; only affirmation resumes music.
Every stage is configurable; defaults (T_a 3 min, T_f 20 s, T_c 90 s, T_r 30 s, T_l 5 min, T_h 10 min) are examples only.

FIG. 4



420

Closed-loop verification 420

Subscribe to player-state; confirm directive took effect within latency bound L; on failure, escalate to next adapter

(e.g., app-remote pause fails -> OS interruption -> device mute). Log directive, acknowledgement, and latency to event log 132.

Graded: AFFIRMED -> full volume; AMBIGUOUS -> duck + prompt; LAPSE -> fade; WITHDRAWN -> pause + cooldown track

FIG. 5

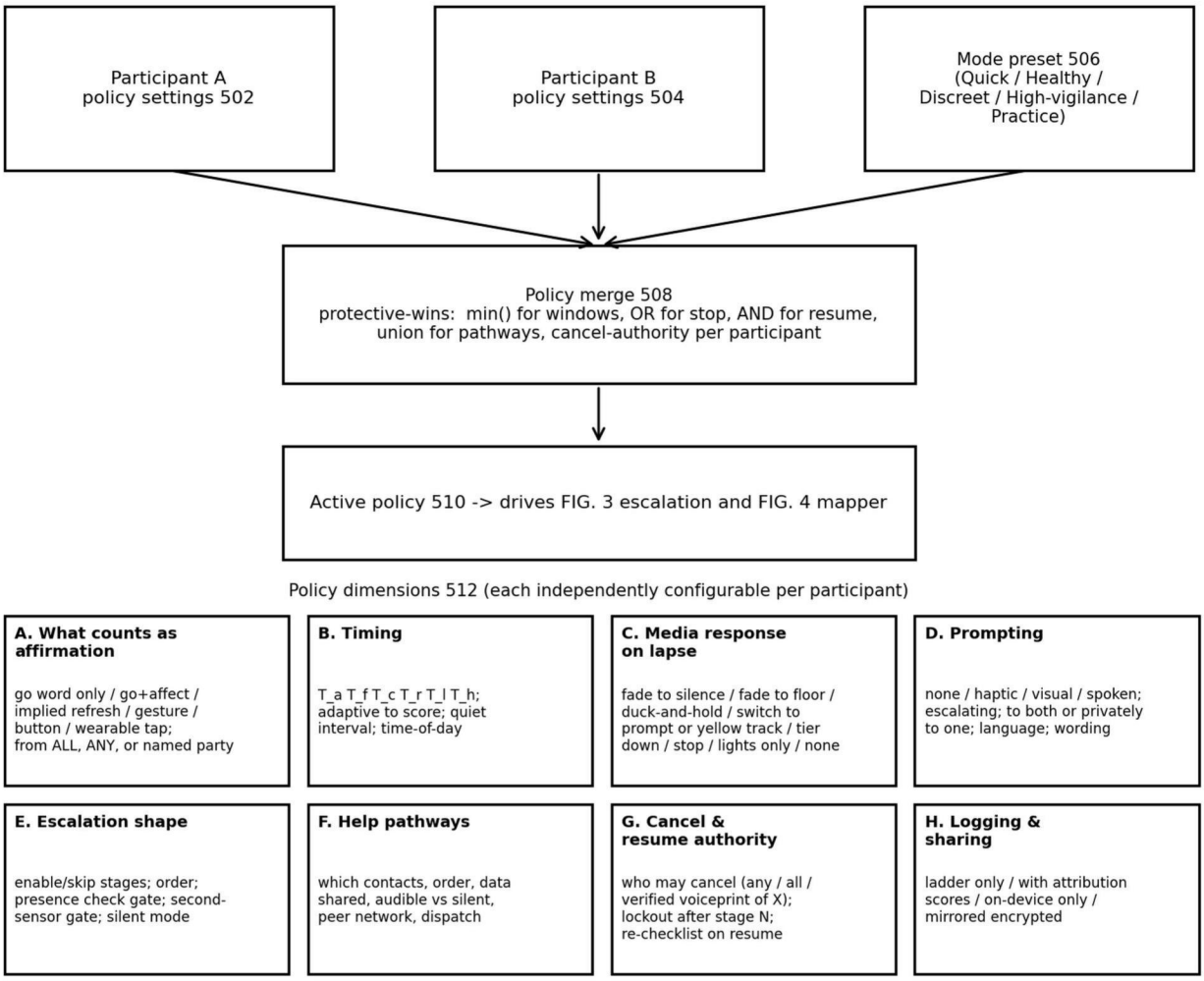
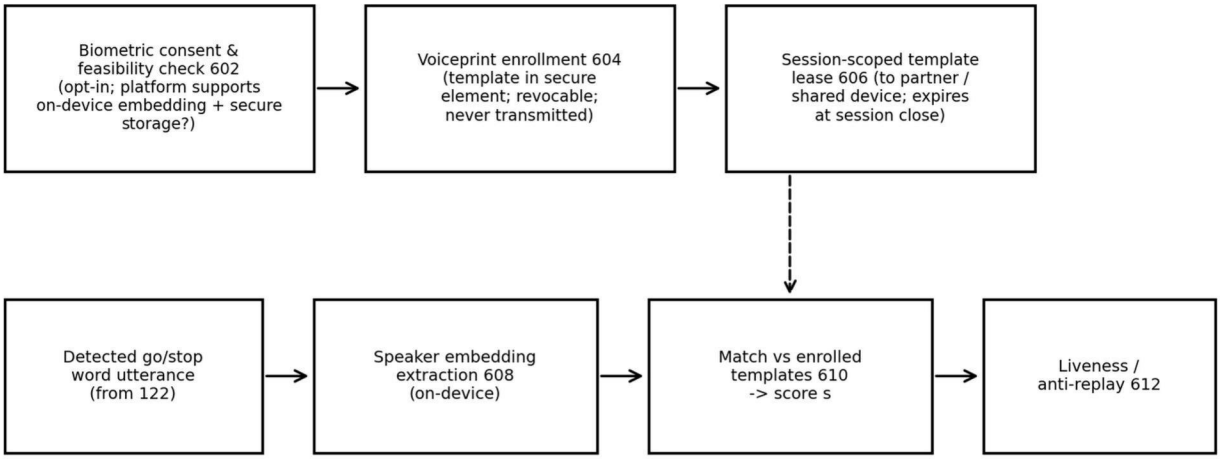

Participant A
policy settings 502
Participant B
policy settings 504
Mode preset 506
(Quick / Healthy /
Discreet / High-vigilance /
Practice)
Policy merge 508
protective-wins: min() for windows, OR for stop, AND for resume,
union for pathways, cancel-authority per participant
Active policy 510 -> drives FIG. 3 escalation and FIG. 4 mapper
Policy dimensions 512 (each independently configurable per participant)
A. What counts as affirmation
go word only / go+affect / implied refresh / gesture / button / wearable tap; from ALL, ANY, or named party
B. Timing
T_a T_f T_c T_r T_l T_h; adaptive to score; quiet interval; time-of-day
C. Media response on lapse
fade to silence / fade to floor / duck-and-hold / switch to prompt or yellow track / tier down / stop / lights only / none
D. Prompting
none / haptic / visual / spoken; escalating; to both or privately to one; language; wording
E. Escalation shape
enable/skip stages; order; presence check gate; second-sensor gate; silent mode
F. Help pathways
which contacts, order, data shared, audible vs silent, peer network, dispatch
G. Cancel & resume authority
who may cancel (any / all / verified voiceprint of X); lockout after stage N; re-checklist on resume
H. Logging & sharing
ladder only / with attribution scores / on-device only / mirrored encrypted

FIG. 6



614

Attribution level assignment 614

```
VERIFIED      s >= tau_v AND liveness pass         -> affirmation credited to that participant
PROBABLE      diarization cluster / proximity only  -> credited only if policy permits PROBABLE
UNATTRIBUTED  no enrolled template or s < tau_p     -> credited only in joint-only mode
CHALLENGE     tau_p <= s < tau_v  -> system asks for a random challenge word, re-verifies
STOP words: always honored at any attribution level; attribution is recorded, never required
```

Each affirmation logged as {pseudonymous participant id, attribution level, score band, liveness result, timestamp} in hash-chained log 132

FIG. 7

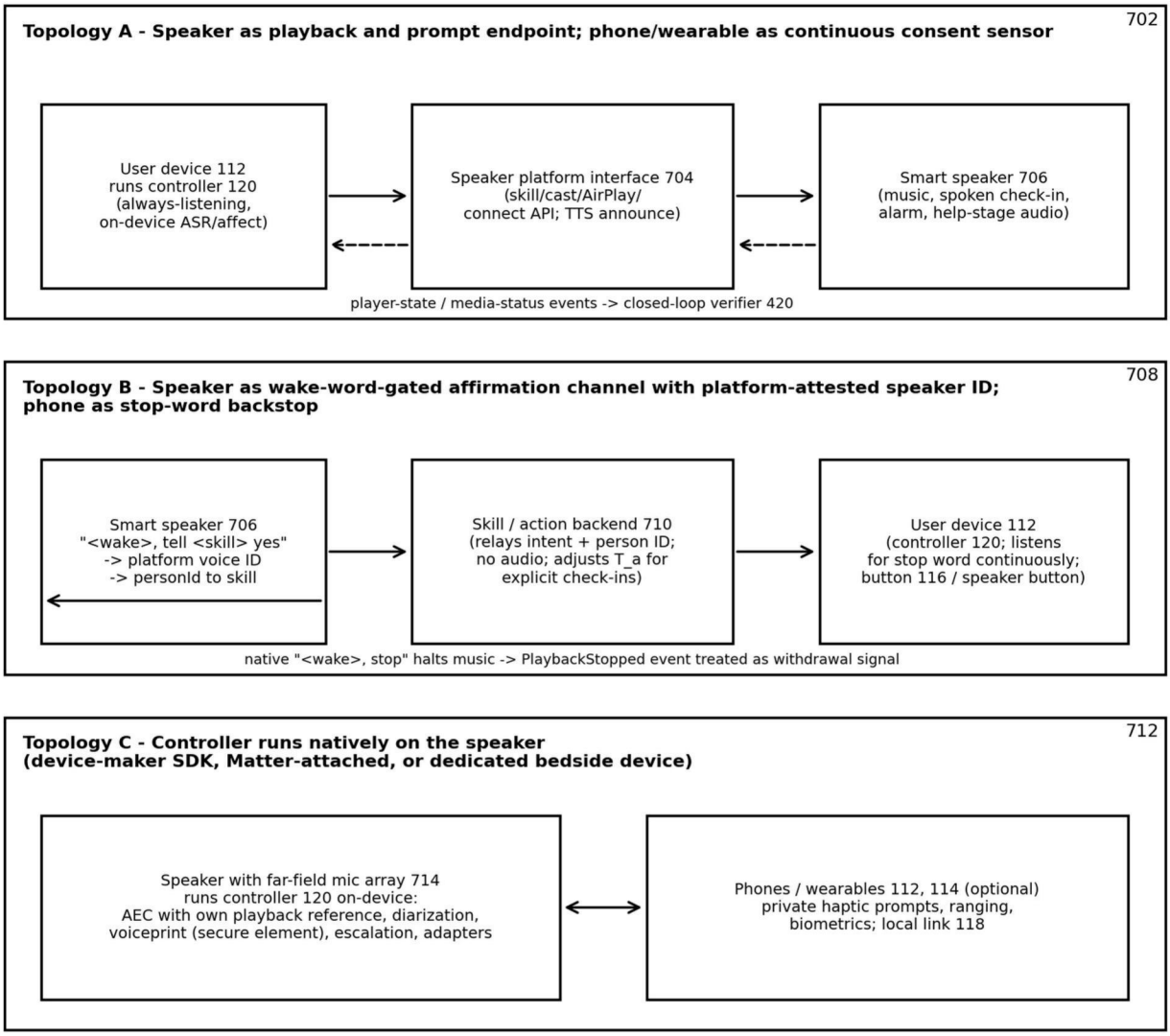
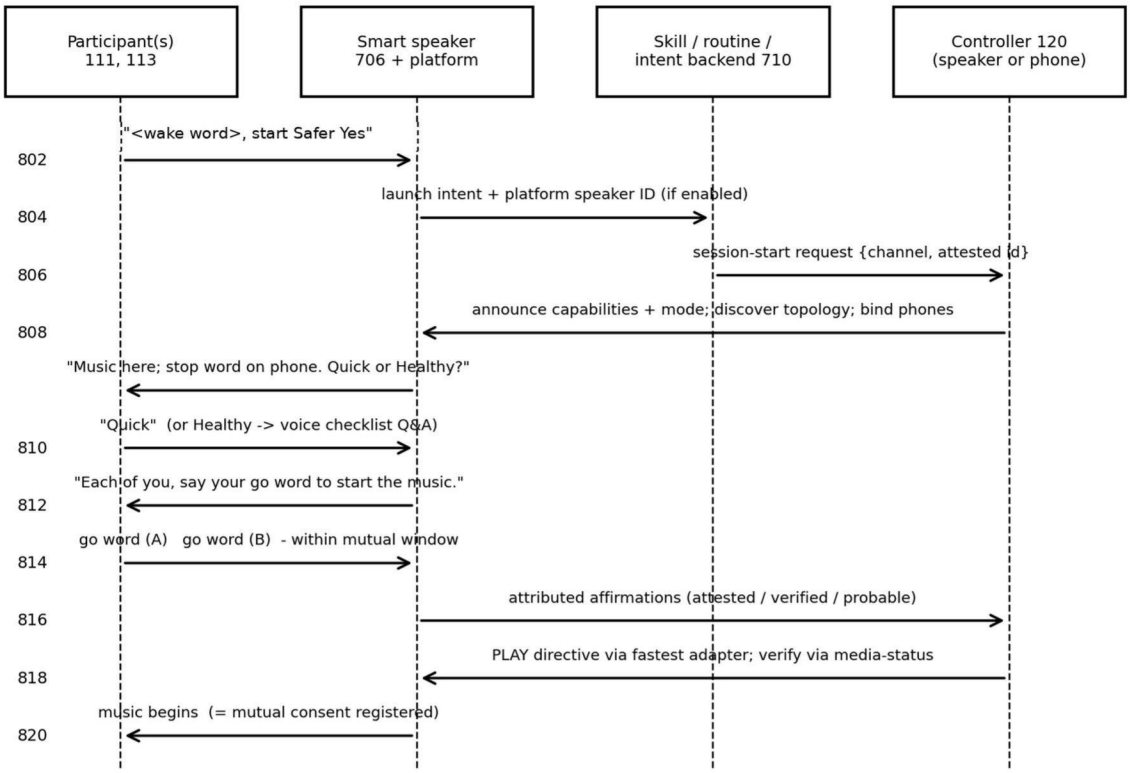

Topology A - Speaker as playback and prompt endpoint; phone/wearable as continuous consent sensor
702
User device 112
runs controller 120
(always-listening,
on-device ASR/affect)
Speaker platform interface 704
(skill/cast/AirPlay/
connect API; TTS announce)
Smart speaker 706
(music, spoken check-in,
alarm, help-stage audio)
player-state / media-status events -> closed-loop verifier 420
Topology B - Speaker as wake-word-gated affirmation channel with platform-attested speaker ID;
phone as stop-word backstop
708
Smart speaker 706
"<wake>, tell <skill> yes"
-> platform voice ID
-> personId to skill
Skill / action backend 710
(relays intent + person ID;
no audio; adjusts T_a for
explicit check-ins)
User device 112
(controller 120; listens
for stop word continuously;
button 116 / speaker button)
native "<wake>, stop" halts music -> PlaybackStopped event treated as withdrawal signal
Topology C - Controller runs natively on the speaker
(device-maker SDK, Matter-attached, or dedicated bedside device)
712
Speaker with far-field mic array 714
runs controller 120 on-device:
AEC with own playback reference, diarization,
voiceprint (secure element), escalation, adapters
Phones / wearables 112, 114 (optional)
private haptic prompts, ranging,
biometrics; local link 118

FIG. 8



822

Guards at start 822

- The invoking voice starts ARMED only; music starts only after every participant's go word (R6). One voice cannot consent for two.
- If no stop-word backstop (listening phone or button) is present: announce reduced protection and require explicit acceptance, or refuse.
- Start is logged {invoker attribution, channel, topology, mode}. '301 Instant Arm / Instant Alarm wake-word modes are honored as aliases.
- "<wake word>, end Safer Yes" = close phrase; native "<wake word>, stop" = withdrawal signal; both logged.

FIG. 9

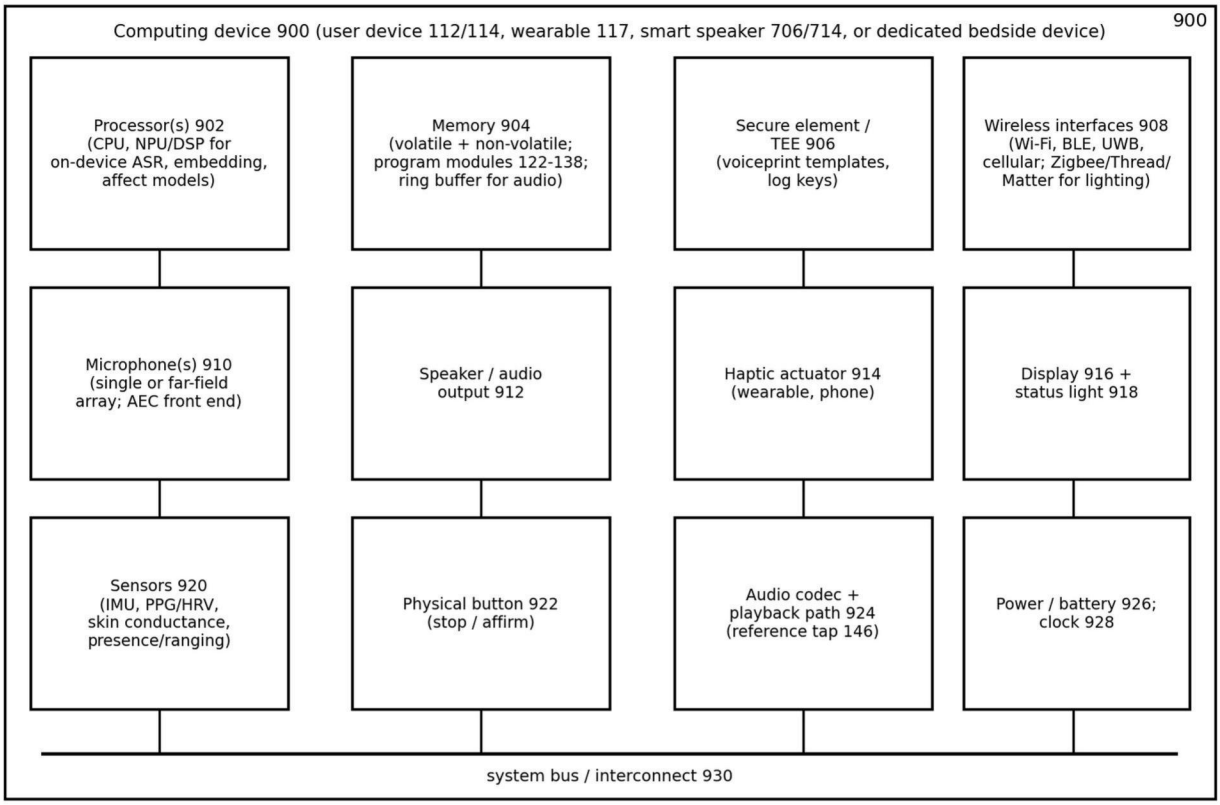
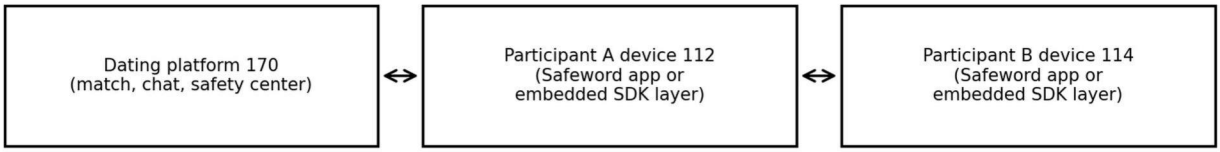

900
Computing device 900 (user device 112/114, wearable 117, smart speaker 706/714, or dedicated bedside device)
Processor(s) 902 (CPU, NPU/DSP for on-device ASR, embedding, affect models)
Memory 904 (volatile + non-volatile; program modules 122-138; ring buffer for audio)
Secure element / TEE 906 (voiceprint templates, log keys)
Wireless interfaces 908 (Wi-Fi, BLE, UWB, cellular; Zigbee/Thread/ Matter for lighting)
Microphone(s) 910 (single or far-field array; AEC front end)
Speaker / audio output 912
Haptic actuator 914 (wearable, phone)
Display 916 + status light 918
Sensors 920 (IMU, PPG/HRV, skin conductance, presence/ranging)
Physical button 922 (stop / affirm)
Audio codec + playback path 924 (reference tap 146)
Power / battery 926; clock 928
system bus / interconnect 930

FIG. 10



1 Match + mutual opt-in 1002

either party proposes a protected date from the chat; both accept; platform passes only a pseudonymous match token

2 Pre-date setup 1004

each on own device: go/stop/close words, policy preset (dating default), trusted contacts, optional voiceprint; platform never receives these

3 Date timer 1006

meeting time/place; check-in timer with escalation to contacts / platform safety partner; location shared only per policy

4 Meeting handshake 1008

proximity (BLE/UWB) or QR between devices binds: match token <-> enrolled participant <-> device in room (identity continuity)

5 Session 1010

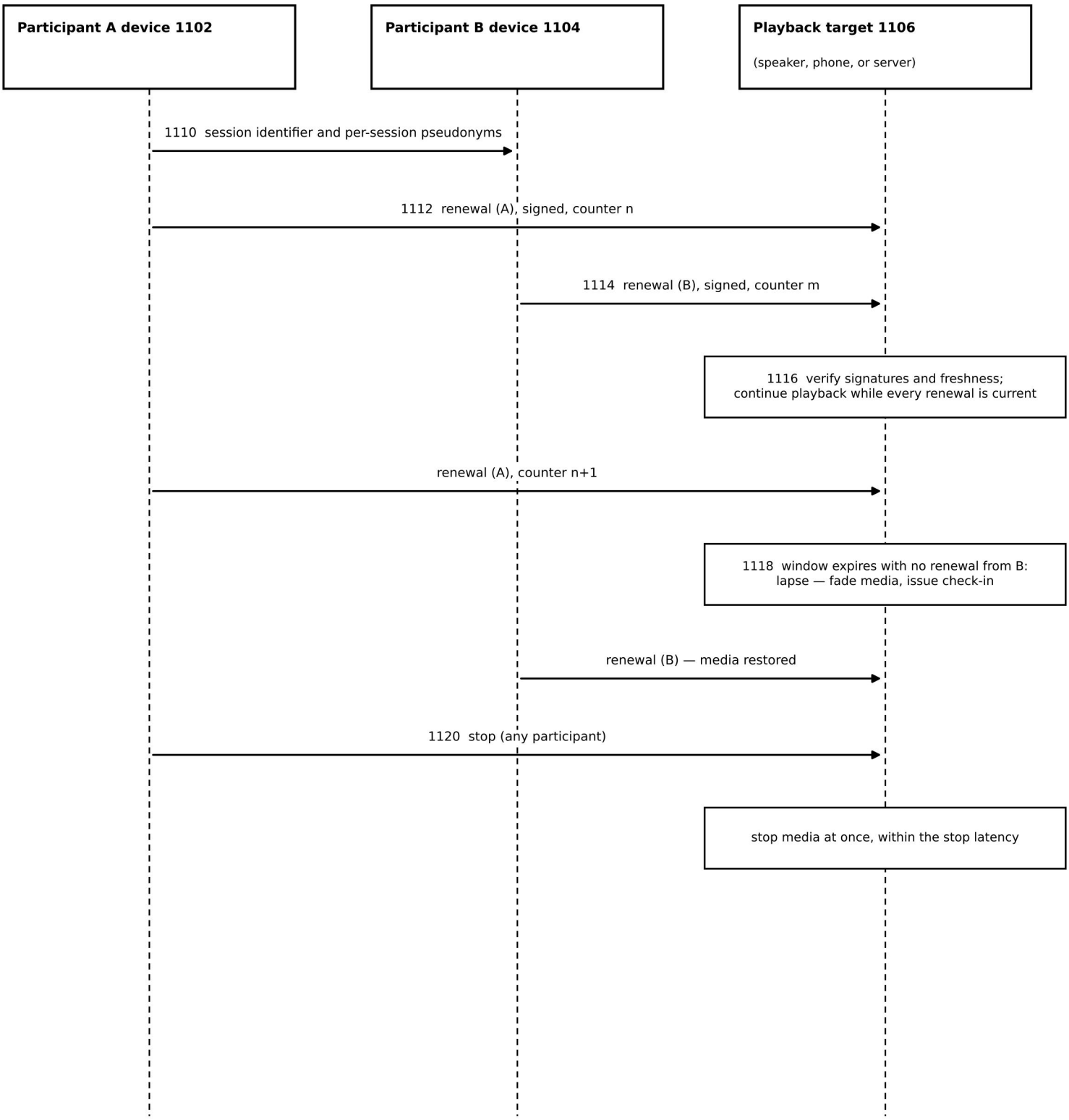
"<wake word>, start Safeword mode" or in-app; affirmation window, lapse, escalation as in FIG. 3; template lease per FIG. 6

6 Close + after 1012

close phrase ends session; platform receives at most {session closed normally | help stage invoked}; log stays with participants

Data minimization 1014: audio, words, consent states, and voiceprints never reach platform 170; unlink at any time; declining has no effect on matching

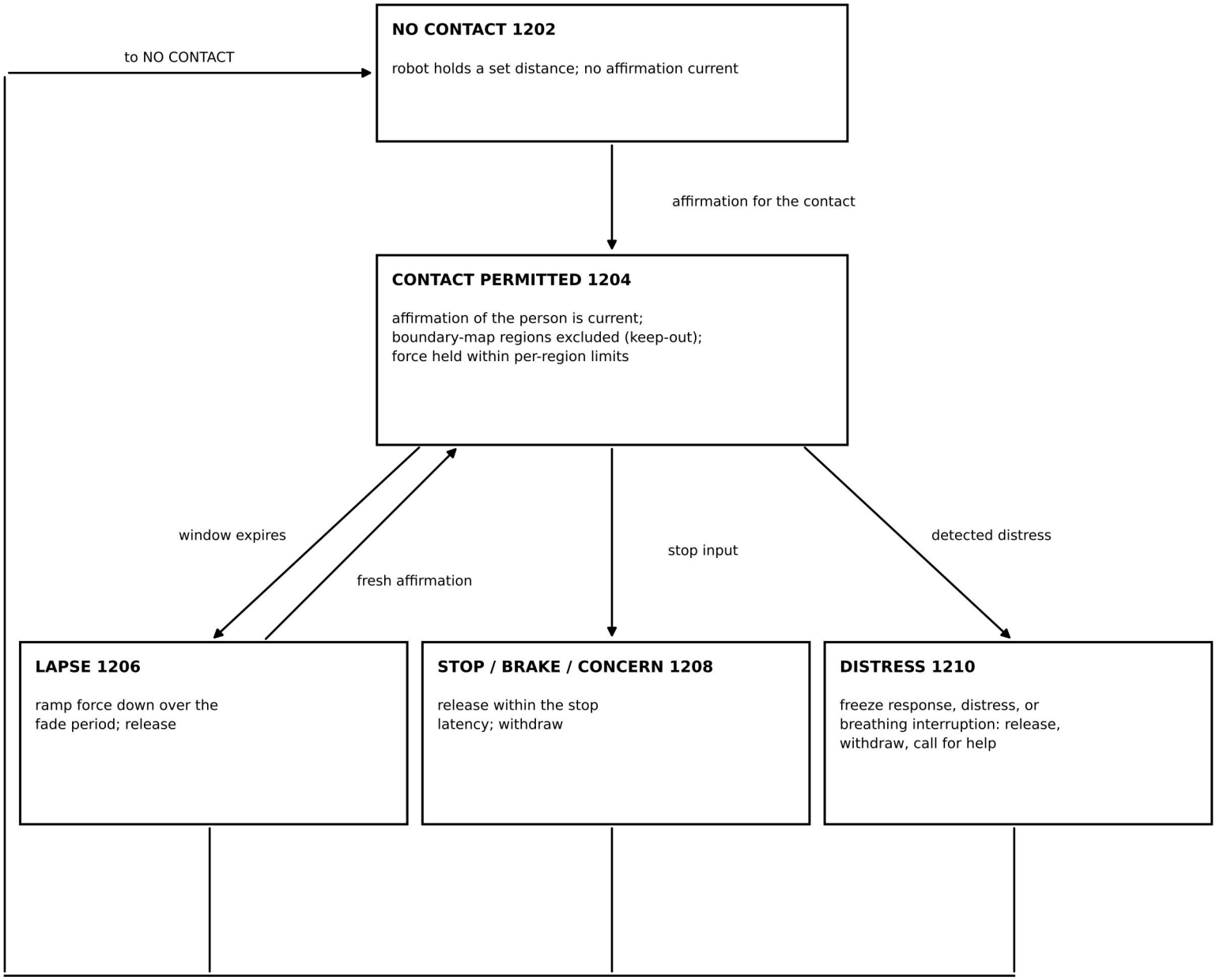
FIG. 11



Renewal-token message 1130

protocol version · session identifier (random, per session) · participant pseudonym (per session) · event type: renewal, stop, brake, concern, interaction, close, help · attribution level · monotonic counter · time stamp · device signature
no audio and no transcript; encrypted to the recipient when leaving the local link; stale or replayed messages are rejected;
a recipient that receives no valid renewal from every participant within the affirmation window enters a lapse (fail-closed)

FIG. 12



Invariants 1212 (every state, every variant)

the robot refuses any command to apply pressure to the neck, whether or not requested;
a stop from the person always releases the person at once;
for companion robots: no mode simulating the overriding of a refusal; no childlike appearance, voice, or persona;
consent sessions only for adults established by age assurance, and only where local law and community rules permit